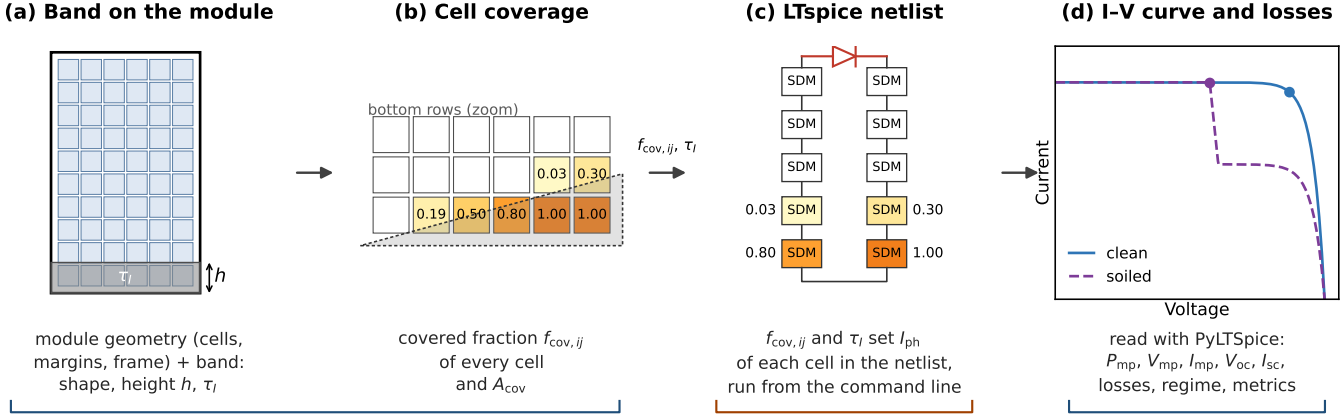
(a) Band on the module
(b) Cell coverage
(c) LTspice netlist
(d) I–V curve and losses
bottom rows (zoom)
0.03
0.30
0.19
0.50
0.80
1.00
1.00
$f_{cov,ij}$, τ_l
τ_l
h
SDM
0.03
0.30
0.80
1.00
Current
Voltage
clean
soiled
module geometry (cells, margins, frame) + band: shape, height h, τ_l
covered fraction $f_{cov,ij}$ of every cell and A_{cov}
$f_{cov,ij}$ and τ_l set I_{ph} of each cell in the netlist, run from the command line
read with PyLTSpice: P_{mp}, V_{mp}, I_{mp}, V_{oc}, I_{sc}, losses, regime, metrics
Python
LTspice
Python